# Calibrated 6D vision for sequential tool manipulation

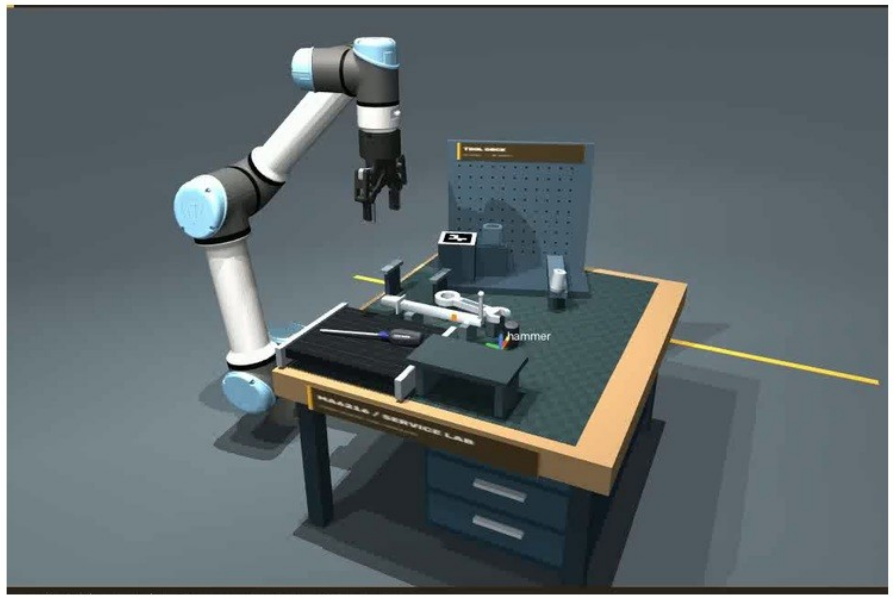


Task demonstration

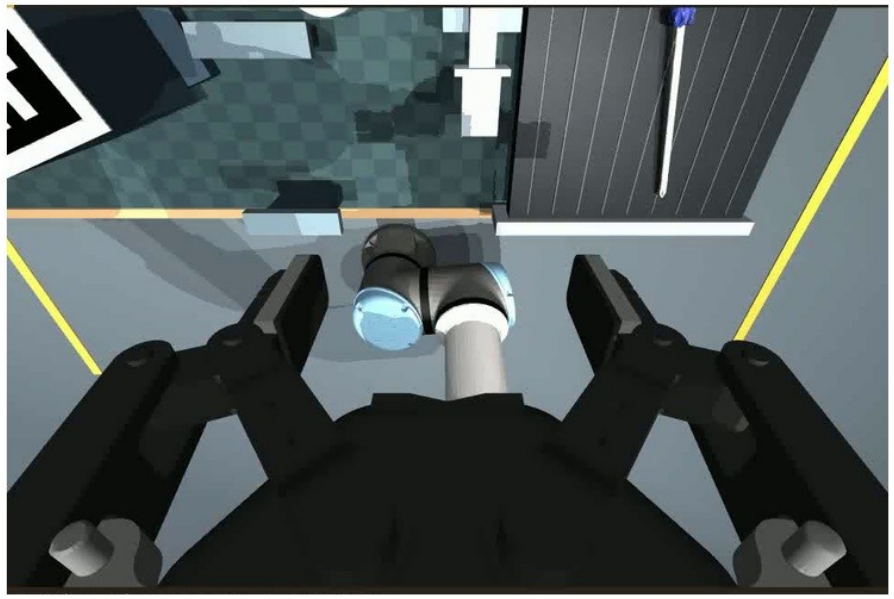
## Vision-guided tool-service sequence

EDITED HIGHLIGHTS | iterative feedback / no disturbance / seed 400

**WORKCELL VIEW**




**WRIST VIEW**



## Initial calibrated workcell

Source playback 000.00 s | 1x playback | Chronological excerpts with cuts

**01 HAMMER RETURN** | 02 MOVING DRIVER | 03 RACK INSTALLATION | 04 REGRASP + HANG

Selected successful run; quantitative results include all 48 attempted trials.

**Play standalone MP4 ↗**

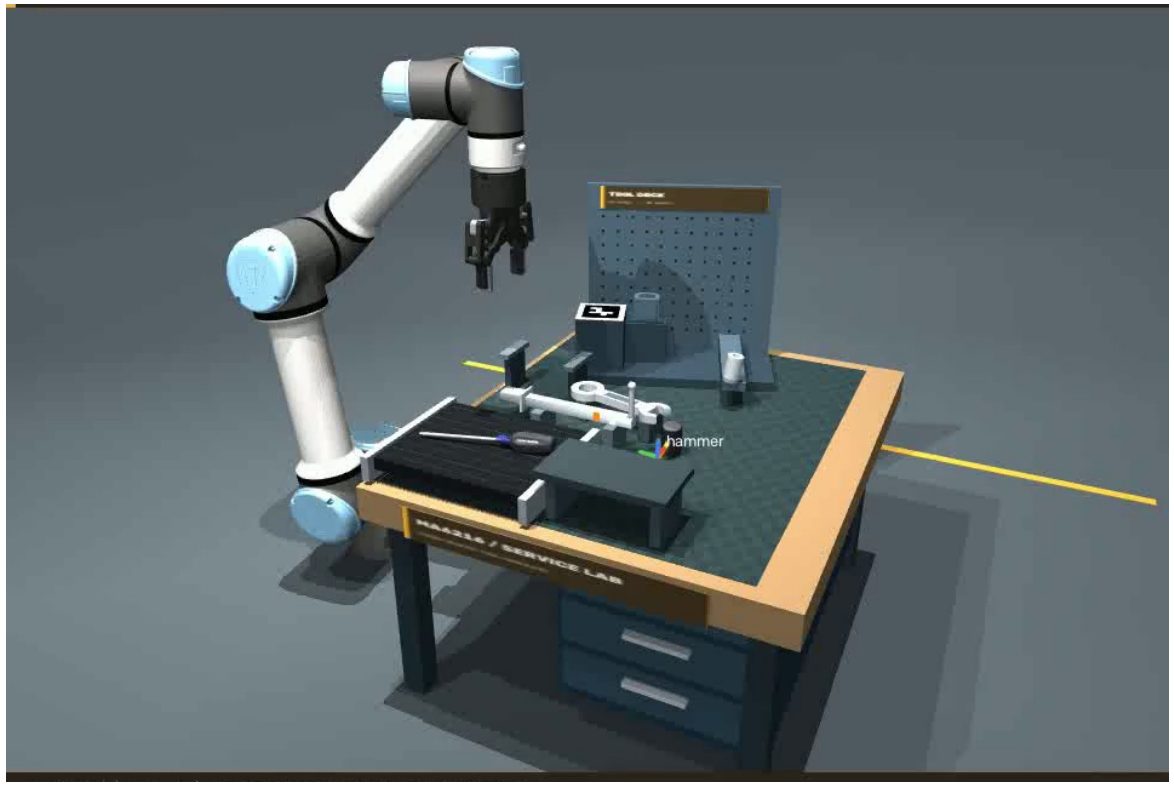
Edited successful run · speed changes and source timestamps labeled

# From visual observation to physical manipulation

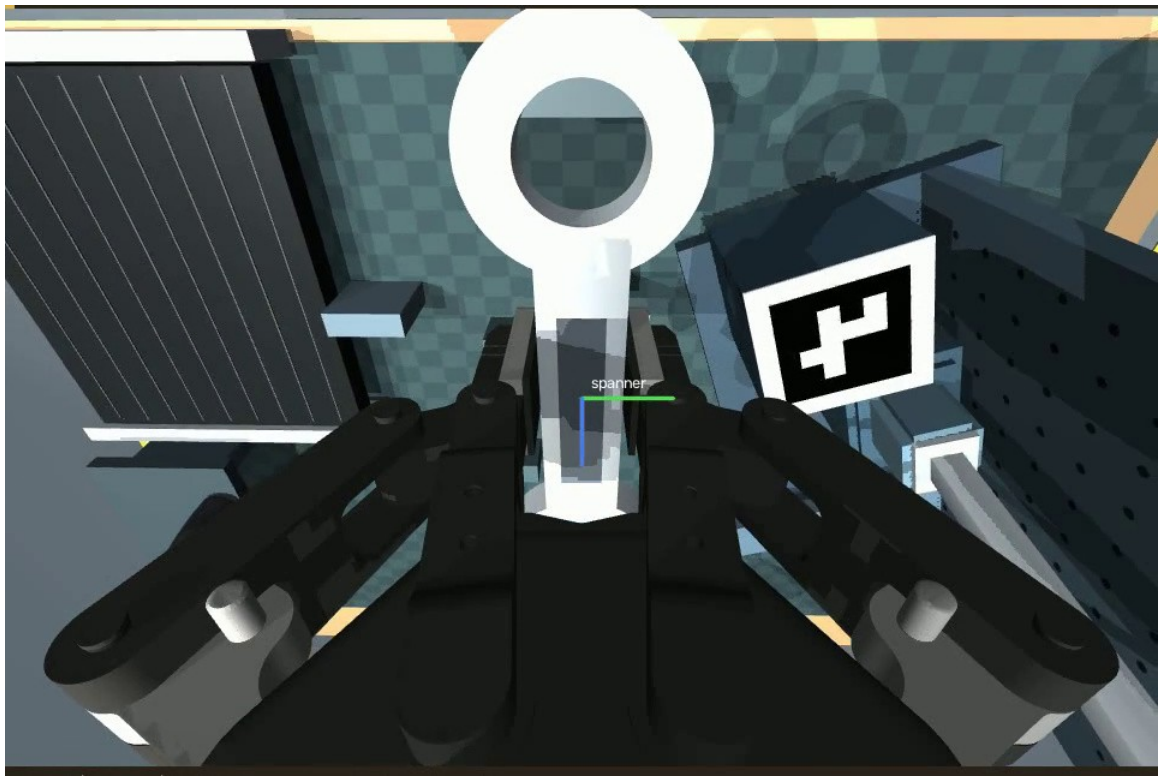


Methods

**RGB-D**
Estimated intrinsics
→
**Localization**
Depth foreground
→
**6D pose**
FoundationPose
→
**Robot target**
Calibrated chain
→
**Execution**
IK + control




**Workcell observation**




**Pose-guided grasp and reobservation**

24 RGB-D reference views → reconstruction → registration / tracking. MegaPose and GraspNet were reviewed, not benchmarked.

# Runtime observation: RGB, metric depth and predicted foreground



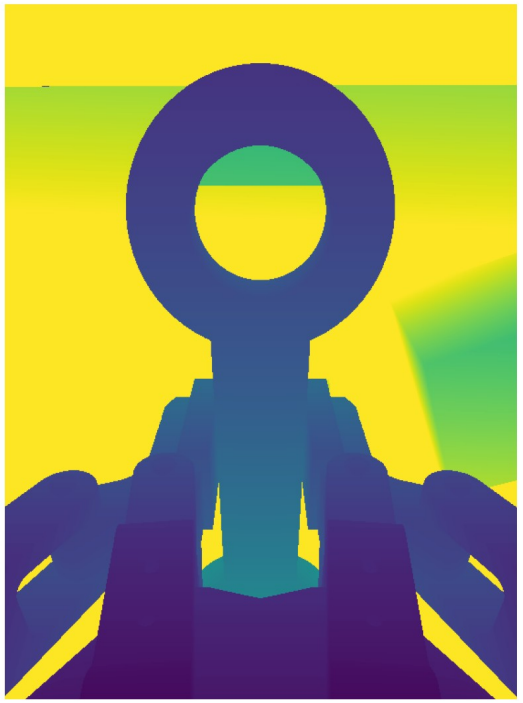
Traceable processing chain

## a Calibrated RGB



Wrist RGB | t = 160.340 s

## b Metric depth




Metric depth (display range clipped)

## c Predicted foreground



Predicted foreground mask

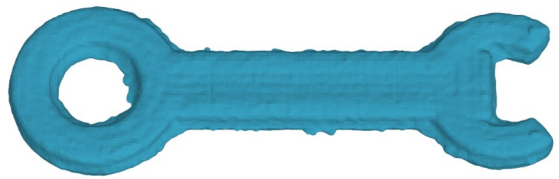
One wrist capture after regrasp: simulation t = 160.340 s, seed 400. Same crop across RGB, depth and predicted mask; no simulator IDs.

# From reconstructed reference to estimated pose and execution



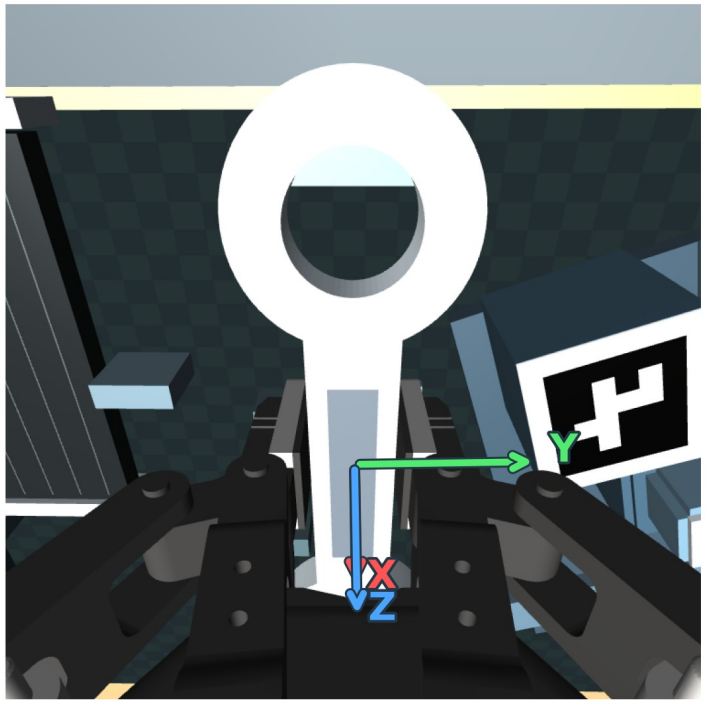
Traceable processing chain

## a Reference mesh



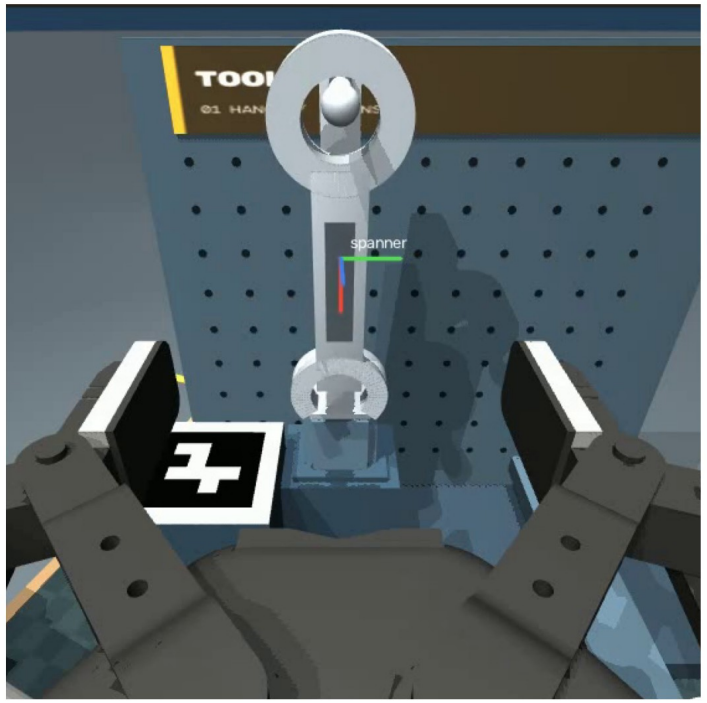
Prior reconstruction | 24 views

## b Estimated pose




Same capture | estimated XYZ axes

## c Released hanging




Later replay frame | t = 195.00 s

The reference is reconstructed beforehand from 24 views. Pose uses the preceding runtime observation; hanging is a later state of the same trial.

# Image-based calibration and held-out validation





**a** Checkerboard observations

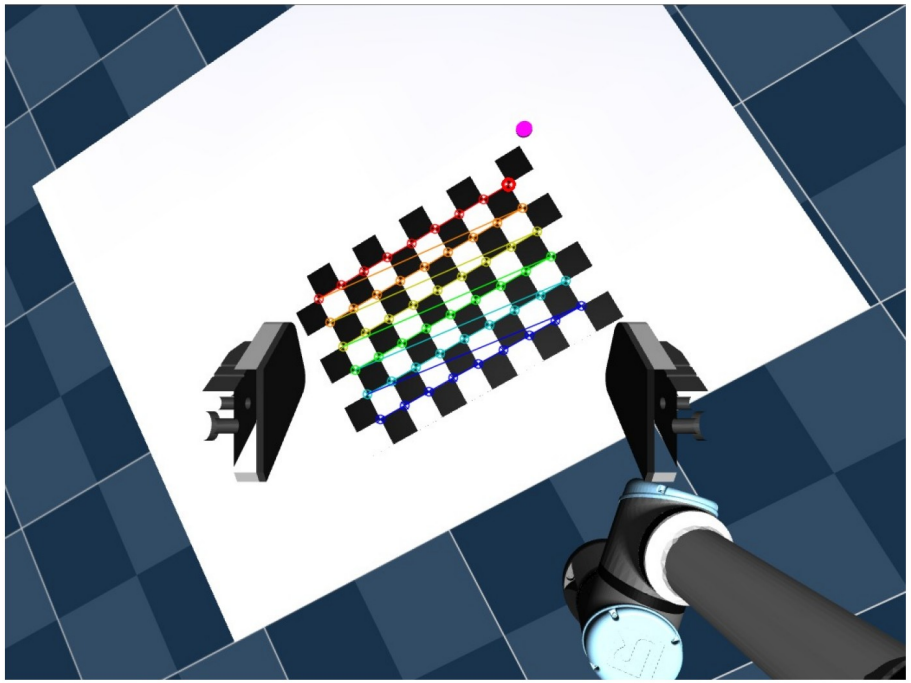
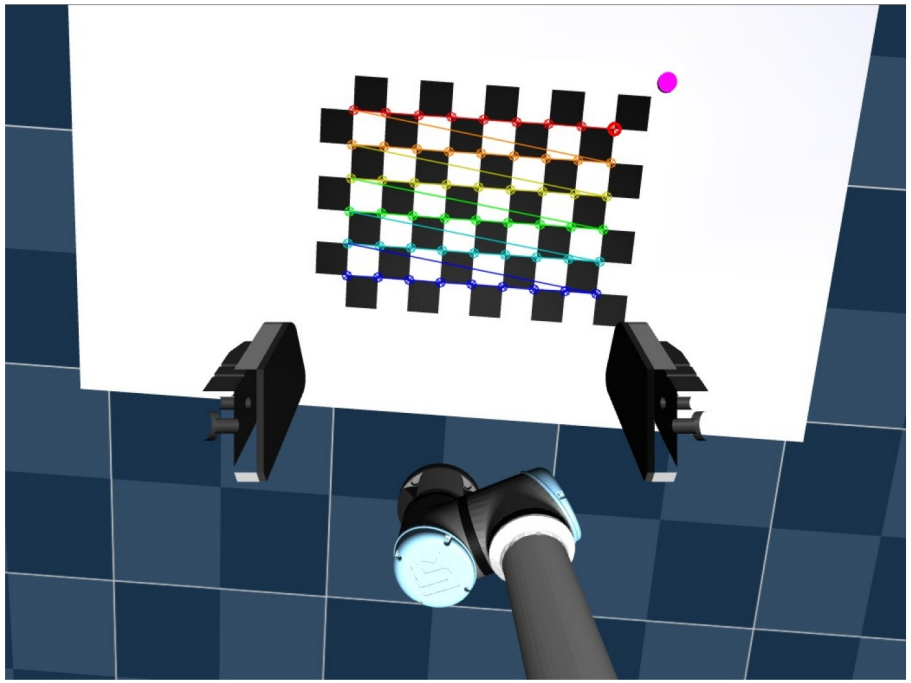


Image-based corner detection

**b** Held-out hand–eye view



PnP + robot FK; frozen validation chain

| Wrist intrinsics | Hand–eye transform | Frozen-chain reprojection |
|---|---|---|
| 24 / 8 train–validation views | 24 / 10 train–validation views | K and transformations fixed |
| 0.02177 px RMSE | 0.852 mm / 0.0227° | 0.05497 px |

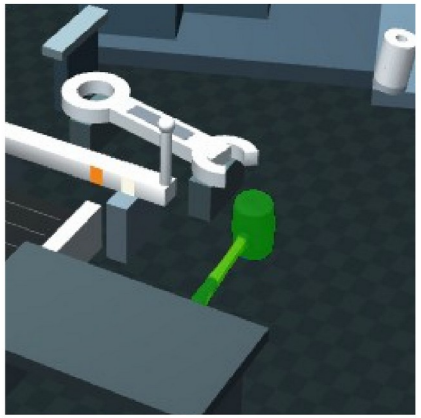
Checkerboard detection + image PnP + robot FK. Simulator parameters are validation references; ideal pinhole images limit transfer.

# Localization: image masks, evaluation unit and assumptions



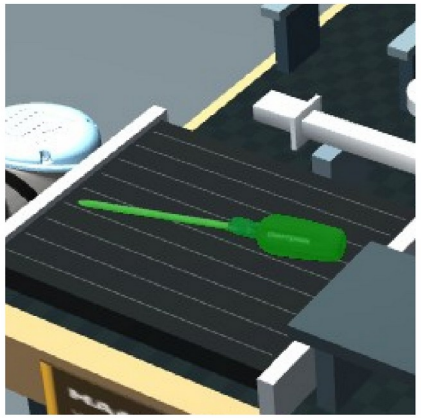


**a** Hammer



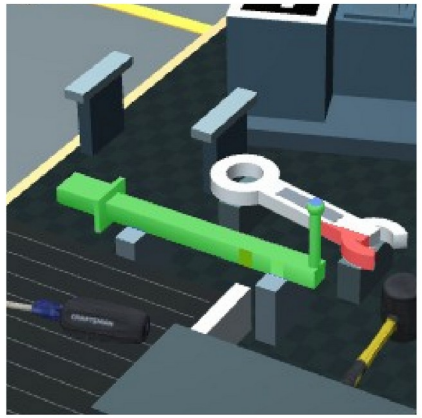
Nominal scene | seed 500

**b** Screwdriver



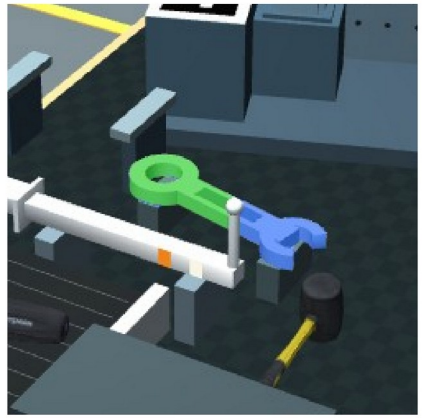
Nominal scene | seed 500

**c** Rack frame



Nominal scene | seed 500

**d** Spanner



Nominal scene | seed 500

| Detector | Independent evaluation | Boundary |
|---|---|---|
| Depth foreground subtraction<br>Known pickup regions | 50 scenes / 200 queries<br>10 scenes per tool–condition cell | Matched empty backgrounds<br>Restricted occlusion / noise |
| Predictions saved first | 160/160 positive queries: IoU ≥ 0.5<br>0/40 absent-target false positives | Wilson and scene-bootstrap intervals<br>Not 200 robot executions |

Nominal seed 500. Green: overlap; blue: missed visible target; red: extra foreground. Crops preserve original overlays.

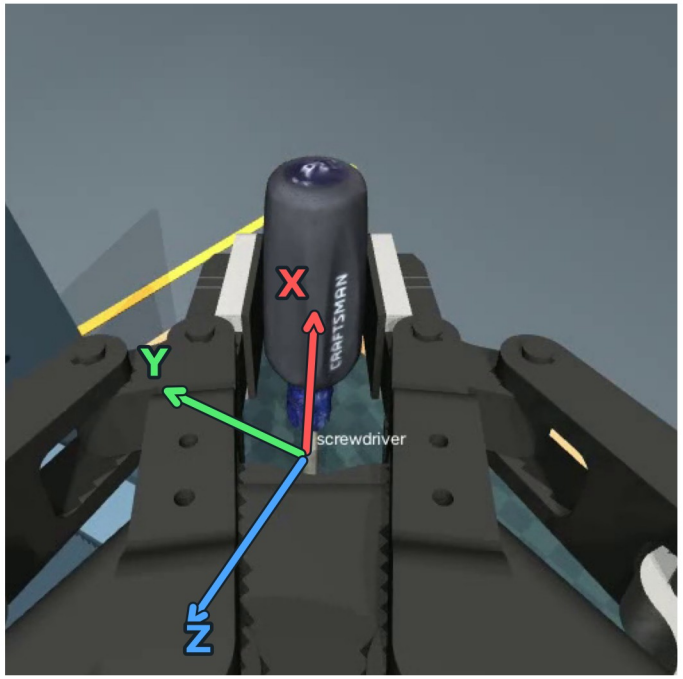
# Estimated object frames guide task-specific targets





## a Screwdriver pose

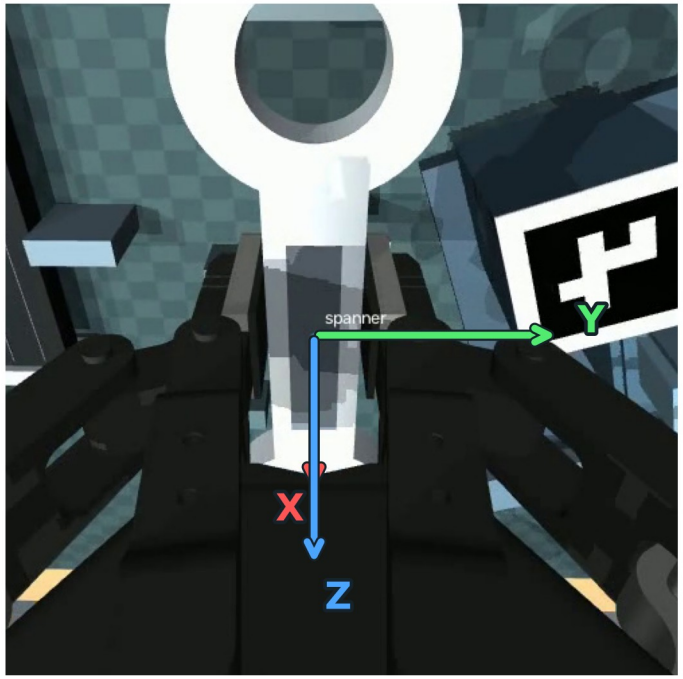
t = 54.40 s | wrist RGB




**Estimated object frame**

## b Spanner pose after regrasp

t = 158.20 s | wrist RGB




**Estimated object frame**

Estimated object frame: X (red), Y (green), Z (blue) | 60 mm display axes | camera perspective retained

$$T_{BO} = T_{BG} \cdot T_{GC} \cdot T_{CO}$$

FK × hand–eye calibration × visual object pose

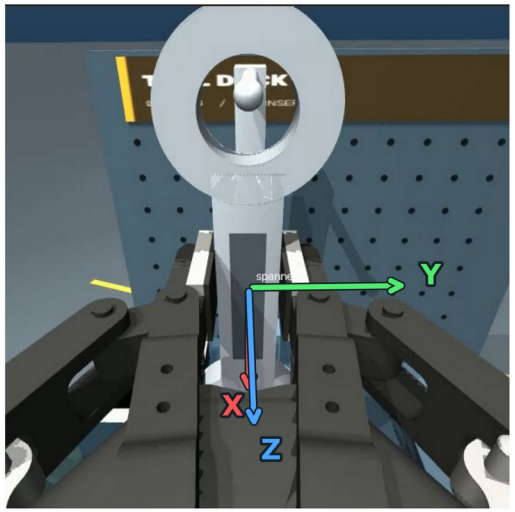
Reobserve after regrasp.

# Visual feedback modes and bounded correction



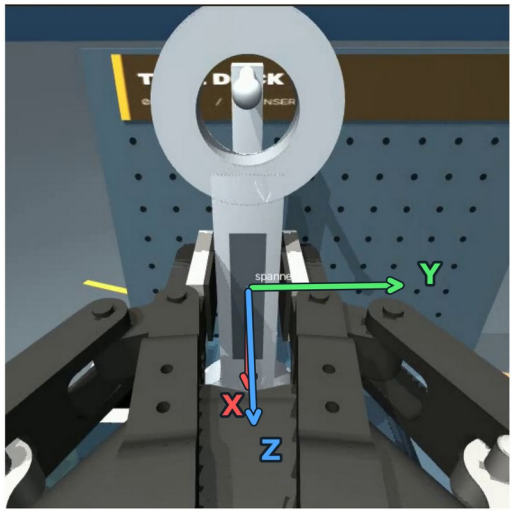


**a Before correction**
t = 171.64 s | wrist RGB




**Visual residual: 1.60°**

**b After reobservation**
t = 172.76 s | wrist RGB




**Visual residual: 0.74°**

Estimated object frame: X (red), Y (green), Z (blue) | 60 mm display axes | camera perspective retained

## Observation schedule

Open loop
Initial acquisition only

Staged
Task-transition observations

Iterative
Repeated precontact correction

## Common control limits

Step cap: 4 mm / 2°
Converged: two observations
within 1.5 mm / 1.5°

Estimated pose residual → bounded correction → reobserve. Same calibration, estimator and guards; discrete precontact control.

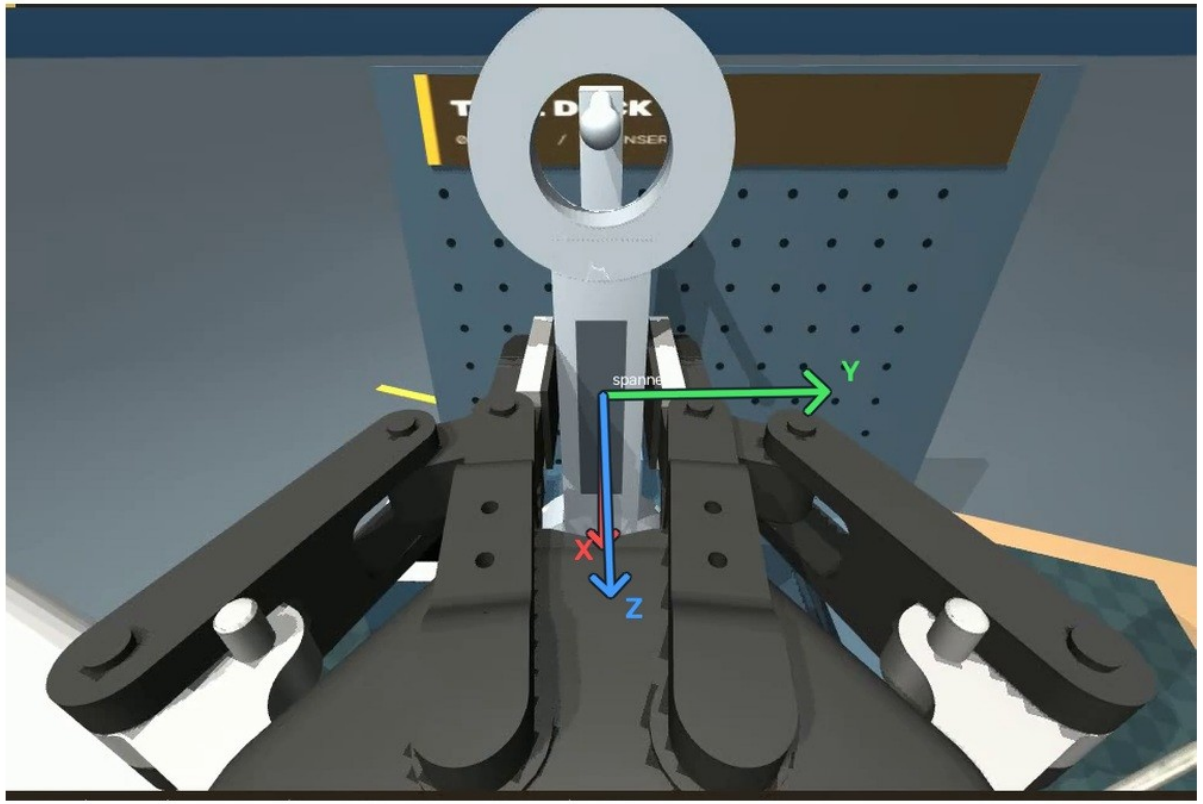
# Logged visual feedback: one correction, three observations





## Visual feedback: observe, correct, observe again

SEPARATE PERTURBED INTEGRATION RUN | 0.125x playback (8x slower) | recorded estimated pose




**OBSERVATION LOG**

**CORRECTING**

Observation 1 / 3

Position residual 1.13 mm

Angular residual 1.60°

Captured: 173.486 s

Available: 173.636 s

**One corrective move commanded**

0.758 mm / 1.037°

**Convergence rule**

Two consecutive observations within 1.5 mm and 1.5°

Source playback 172.125 s | Simulation time 174.127 s | Latest available measurement held between updates

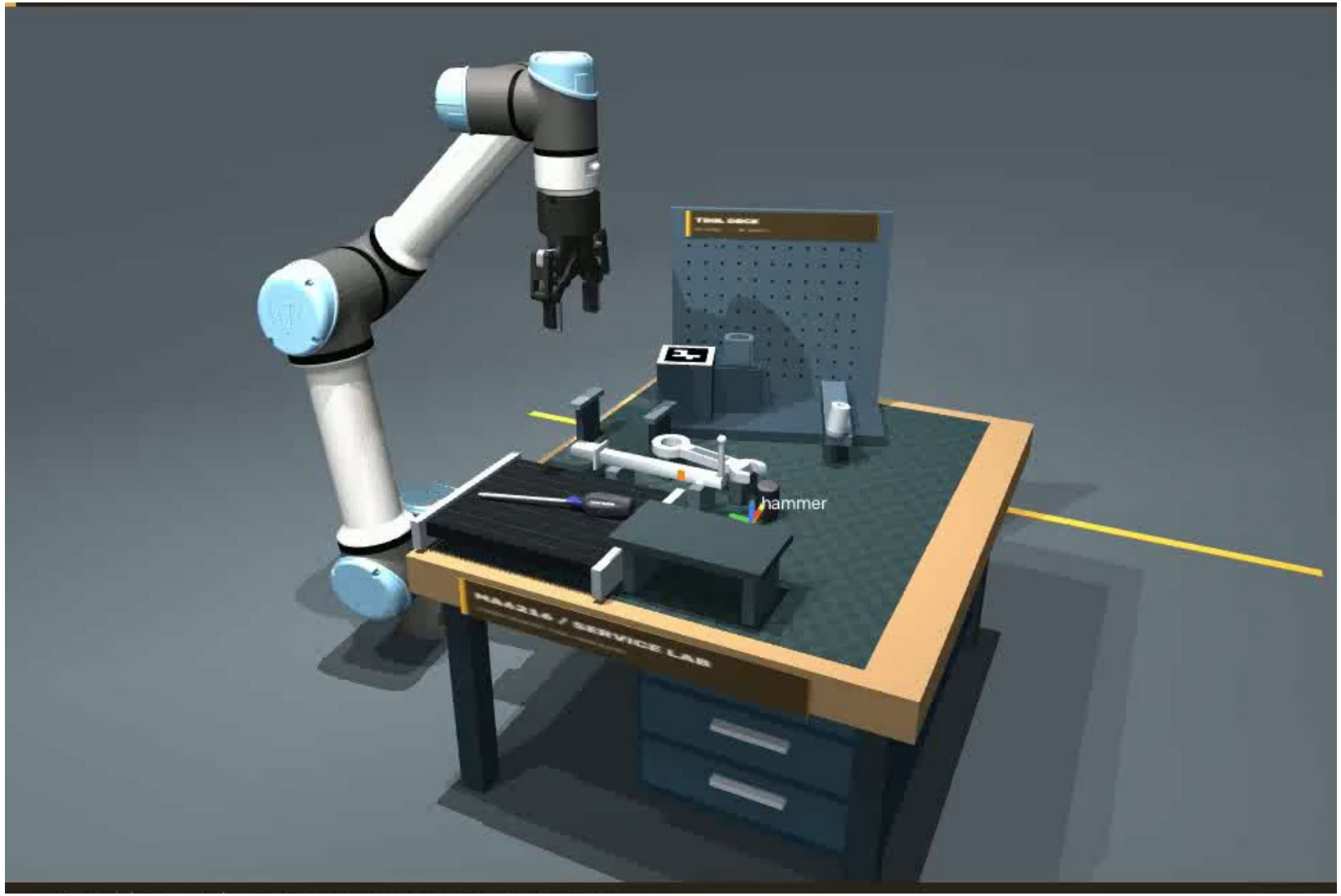
Residuals compare estimated pose with the desired pose; not ground-truth error. This run is not pooled into the 48 trials.

**Play standalone MP4 ↗**

8× slower playback · logged values held between available observations

# Paired evaluation in the complete workcell






Evaluation protocol

## 48 complete-task attempts

3 feedback modes
× 2 disturbance conditions
× 8 paired seeds

## Outcome and uncertainty

All four physical checks must pass.

Wilson 95% intervals
Exact paired tests + Holm
Trial-level continuous summaries

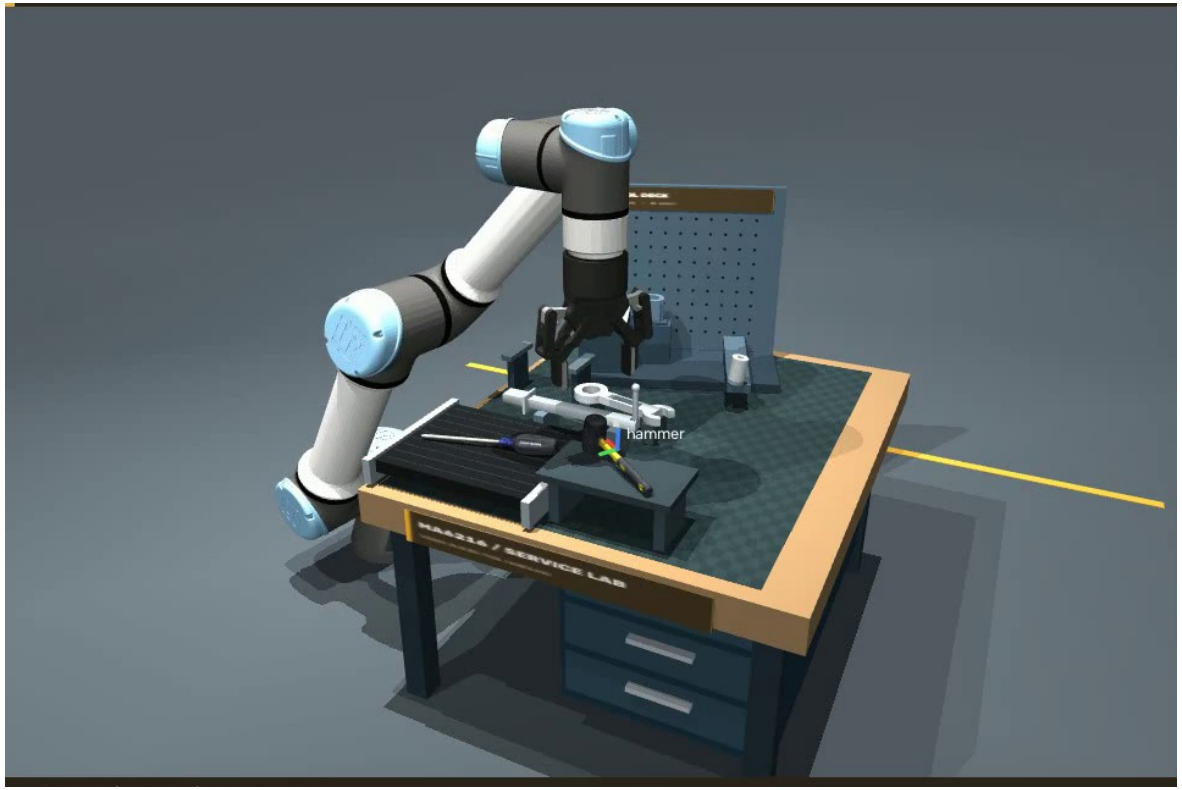
Seeds 400–407 reused.
Paired replication, not a new holdout.
Early failures remain included.

# Physical return and independent rack installation

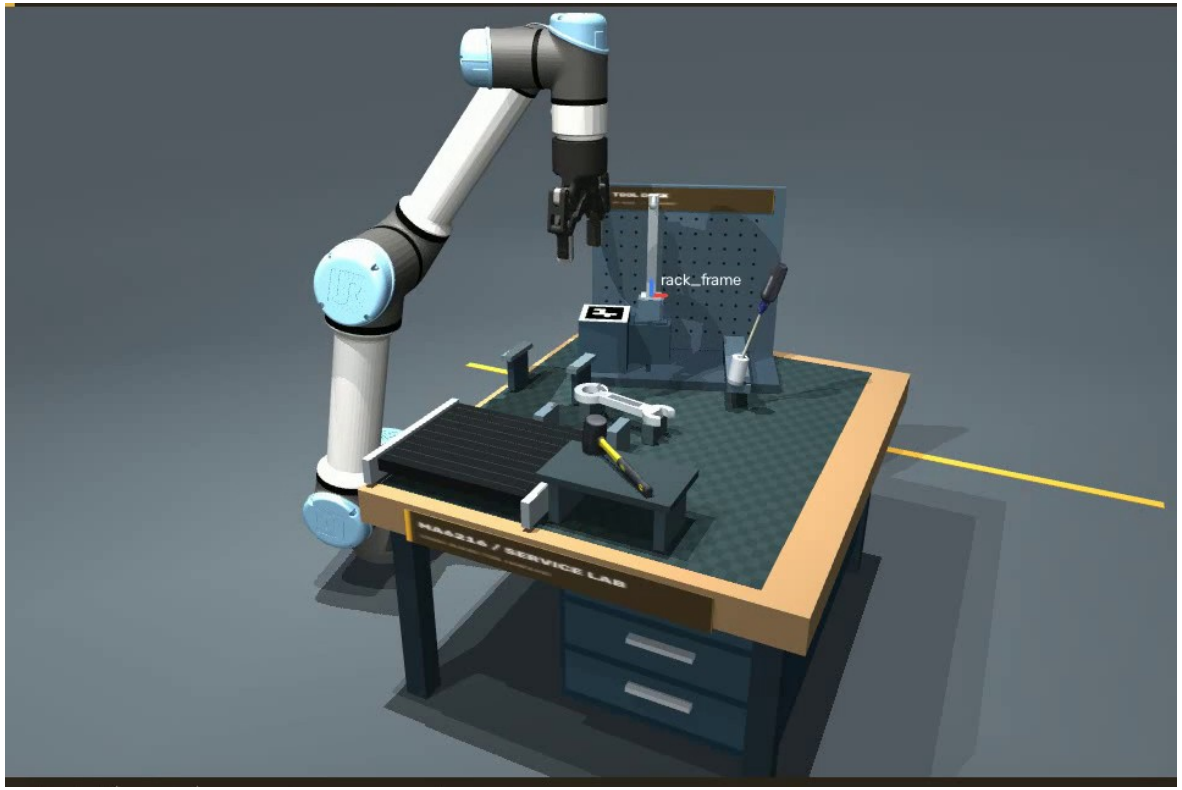


Representative successful run

## a Released hammer placement · 25.00 s




## b Released rack fixture · 134.40 s



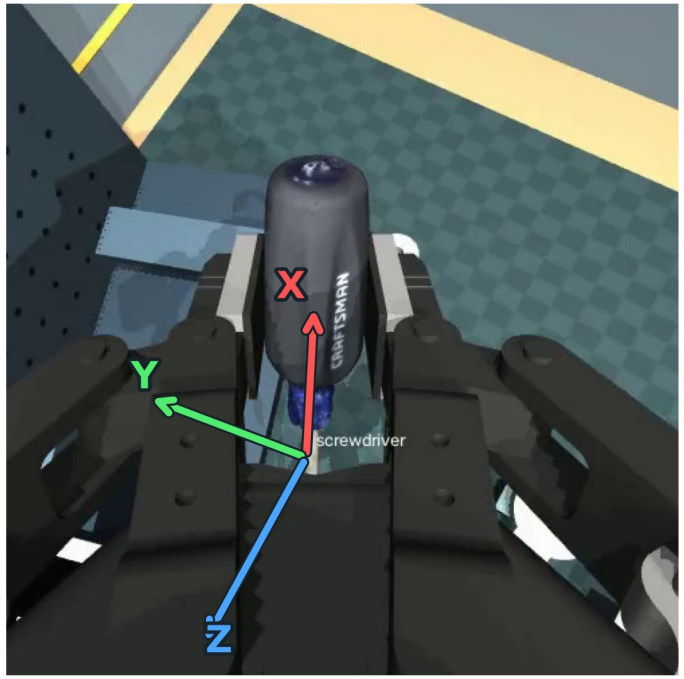

One continuous replay. The rack is installed as an independent object; released support and stability determine acceptance.

# Tilted insertion: orientation, contact and released placement

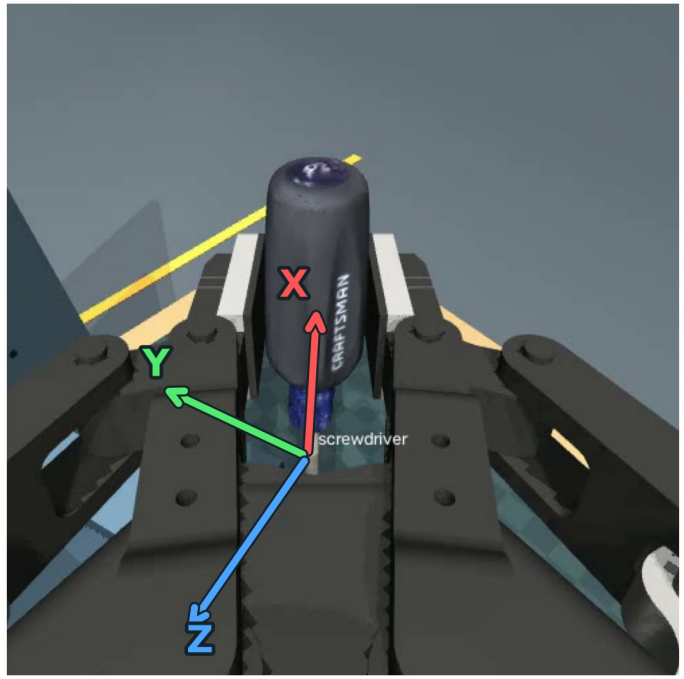


Representative successful run

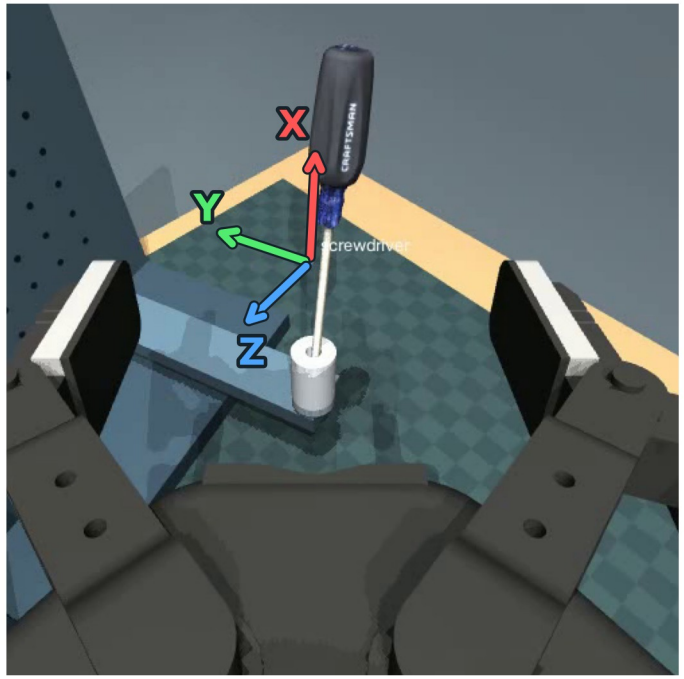
**a Reorient**
t = 50.00 s | wrist RGB




**b Insert**
t = 60.00 s | wrist RGB




**c Release**
t = 70.00 s | wrist RGB




**Tool / gripper** **Insertion region** **Placed tool**

Estimated object frame: X (red), Y (green), Z (blue) | 60 mm display axes | camera perspective retained

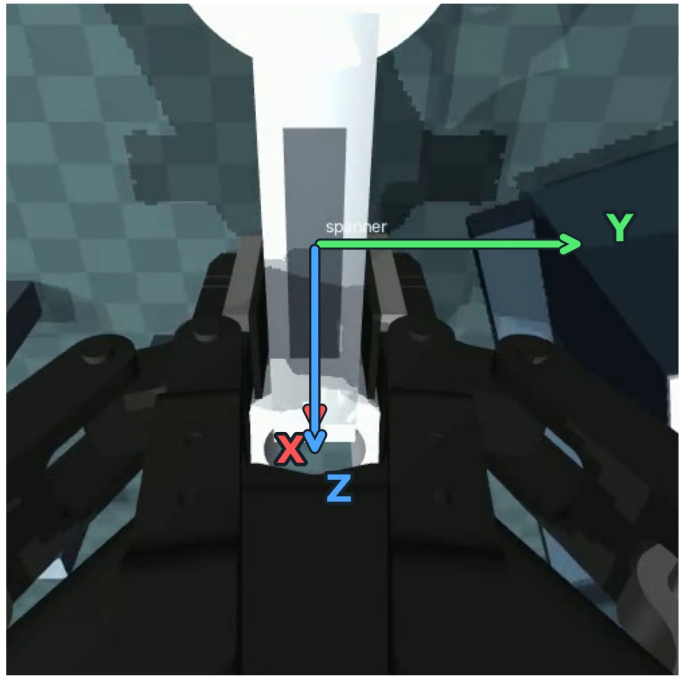
Enlarged panels use matched pixel windows. Contact is partly occluded; physical acceptance comes from the recorded checks.

# Physical regrasp: support, clearance and grasp reacquisition



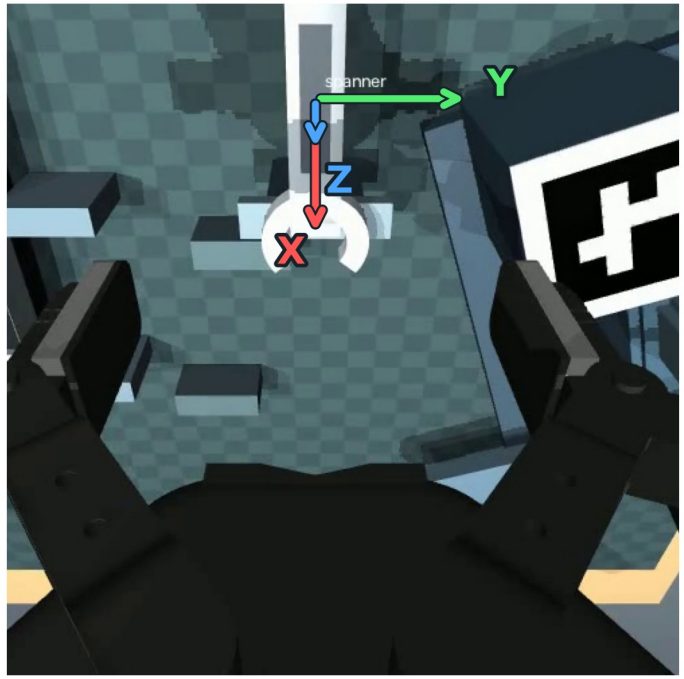
Representative successful run

### a Place on support
t = 147.00 s | wrist RGB



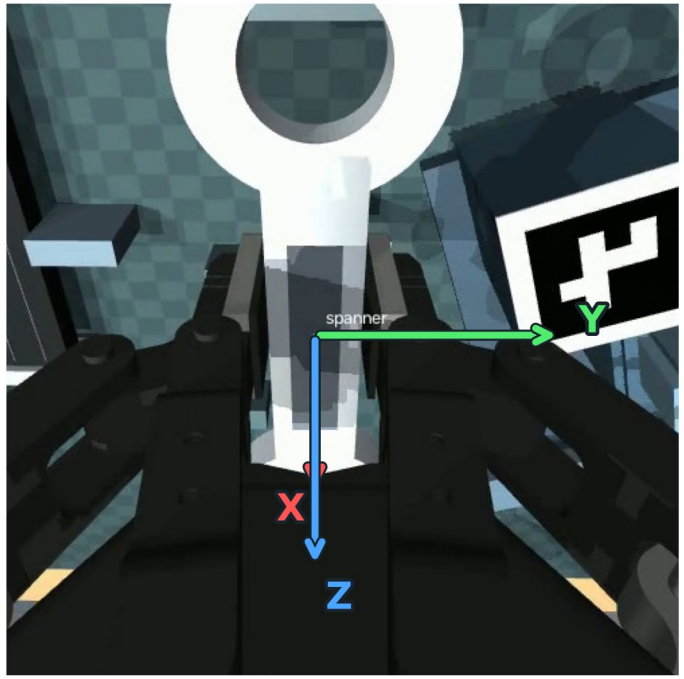

**Original grasp**

### b Clear fingers
t = 150.40 s | wrist RGB




**Open clearance**

### c Acquire new grasp
t = 158.20 s | wrist RGB




**New grasp region**

Estimated object frame: X (red), Y (green), Z (blue) | 60 mm display axes | camera perspective retained

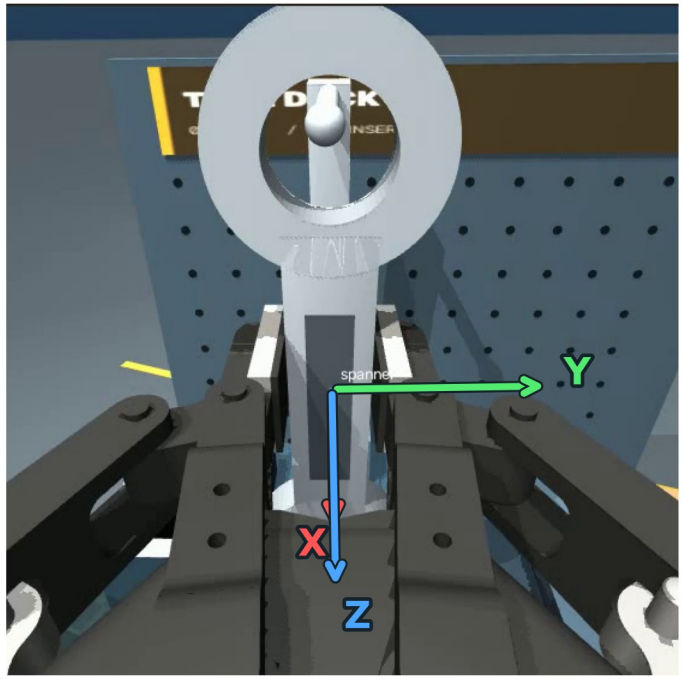
One continuous trajectory. Enlarged views expose the grasp interface; new pose observation precedes reacquisition.

# Hole-over-pin placement: local geometry and final support



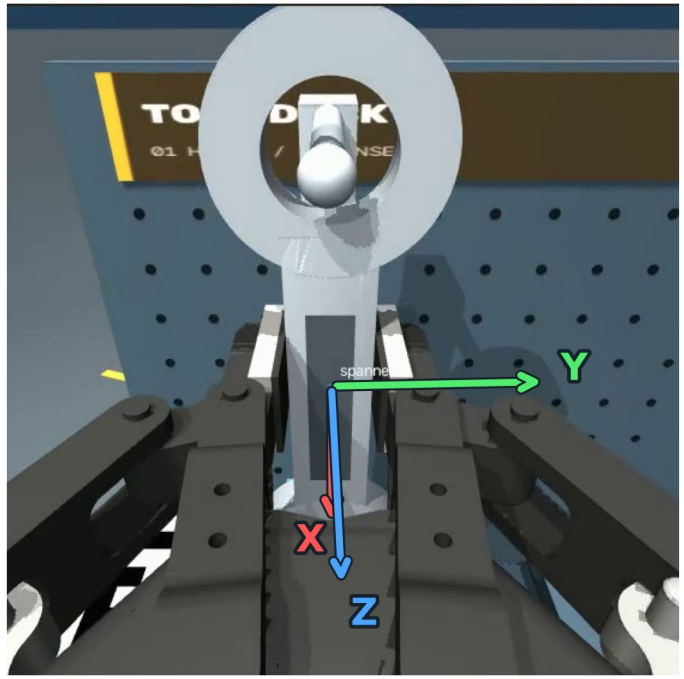
Representative successful run

**a Align**
t = 171.00 s | wrist RGB



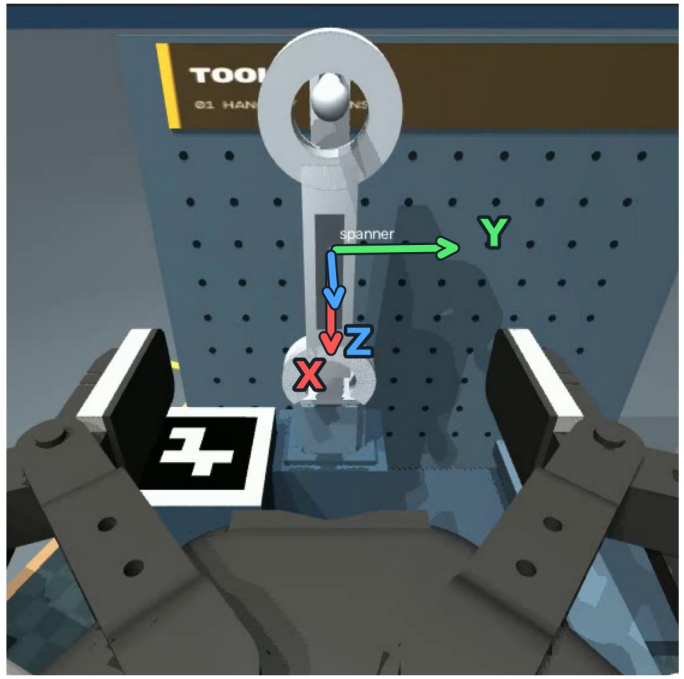

**Ring / pin interface**

**b Thread**
t = 179.00 s | wrist RGB




**Pin inside ring**

**c Release**
t = 195.00 s | wrist RGB




**Released support**

Estimated object frame: X (red), Y (green), Z (blue) | 60 mm display axes | camera perspective retained

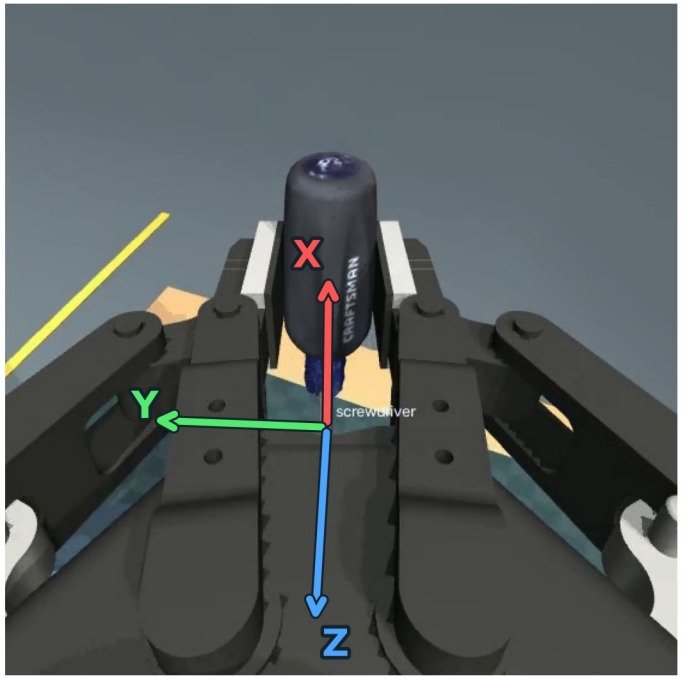
Four terminal checks passed. Separate passive-stability audit: 30 s. Pose axes are visualization overlays, not measured contact forces.

# Matched-seed comparison: insertion stop versus released placement



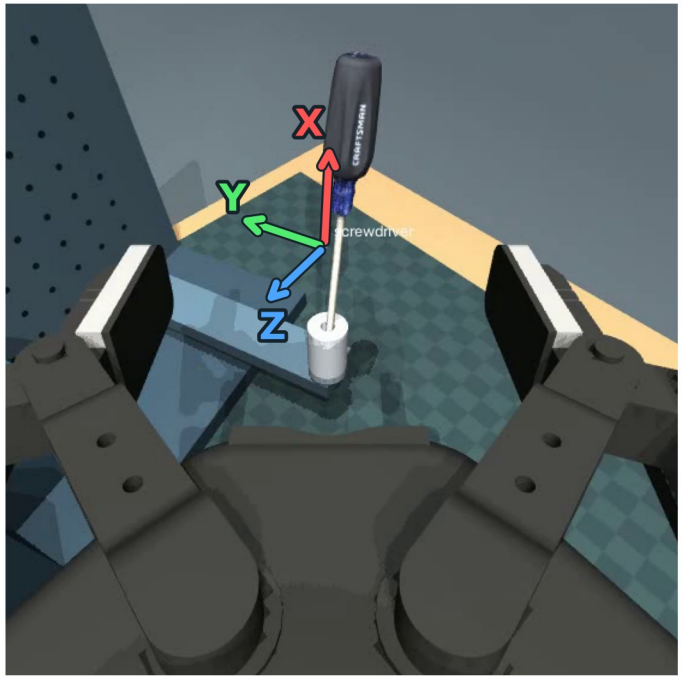
Controlled examples

### a Open loop / seed 400
t = 57.96 s | wrist RGB




**Insertion guard stop**

### b Iterative / seed 400
t = 70.00 s | wrist RGB




**Released placement**

Estimated object frame: X (red), Y (green), Z (blue) | 60 mm display axes | camera perspective retained

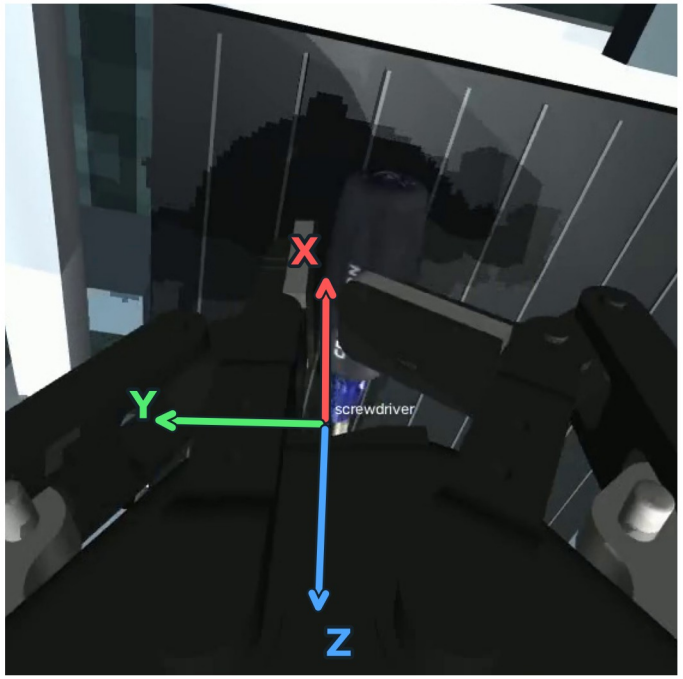
Seed 400, no disturbance; endpoint times differ. A selected paired example does not establish statistical superiority.

# Shared failure boundary: gripper closure before insertion



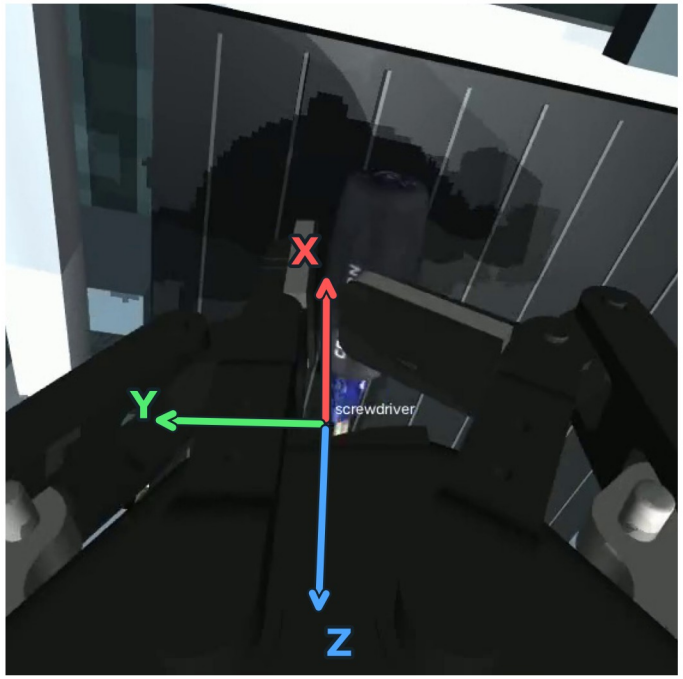
Controlled examples

**a Staged / seed 401**
t = 40.24 s | wrist RGB




**Pickup guard stop**

**b Iterative / seed 401**
t = 40.24 s | wrist RGB




**Same failure stage**

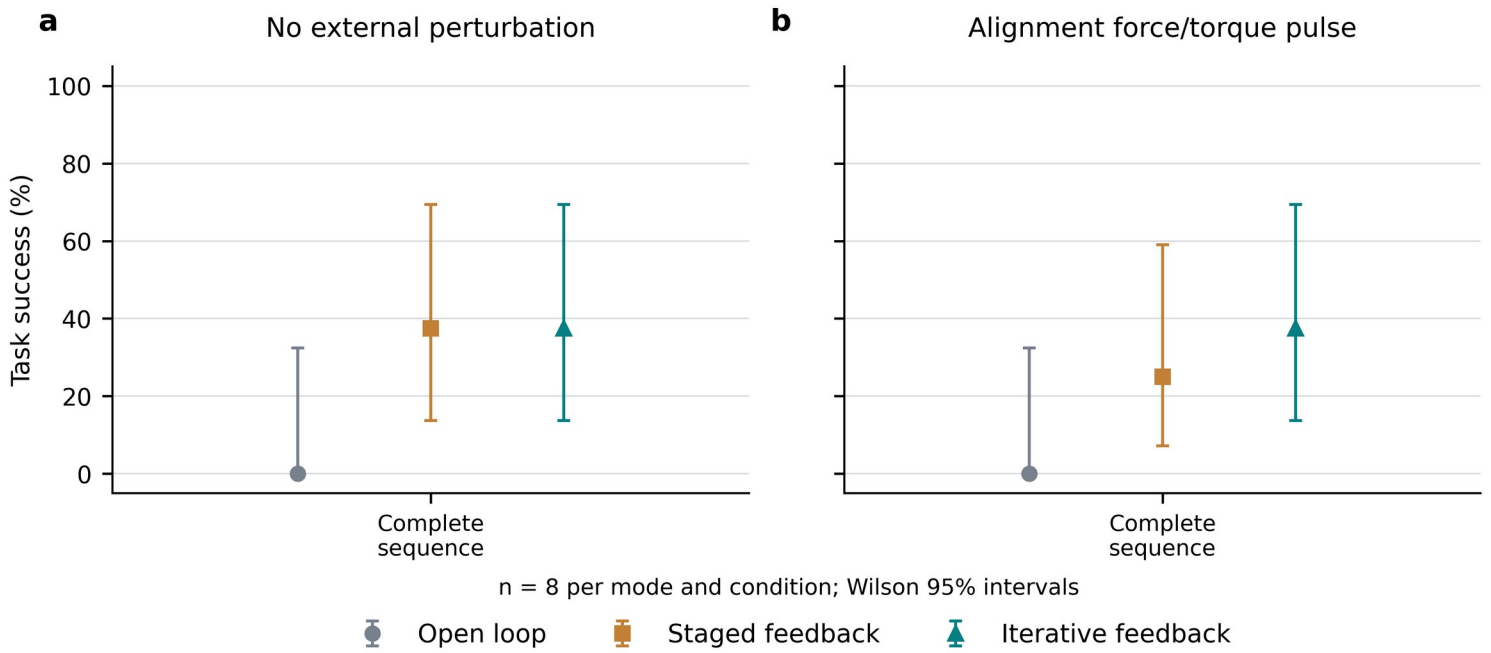
Estimated object frame: X (red), Y (green), Z (blue) | 60 mm display axes | camera perspective retained

Seed 401: both modes stop at gripper closure. Stage and guard attribution are from logs; all failed attempts remain in the analysis.

# Whole-sequence results and statistical uncertainty



Quantitative results




| Mode | None | Pulse |
|---|---|---|
| Open loop | 0/8 | 0/8 |
| Staged | 3/8 | 2/8 |
| Iterative | 3/8 | 3/8 |

11/48 complete successes.
All four adjusted paired p-values: 1.0.
No statistically established superiority or equivalence in this sample.

## Interpretation

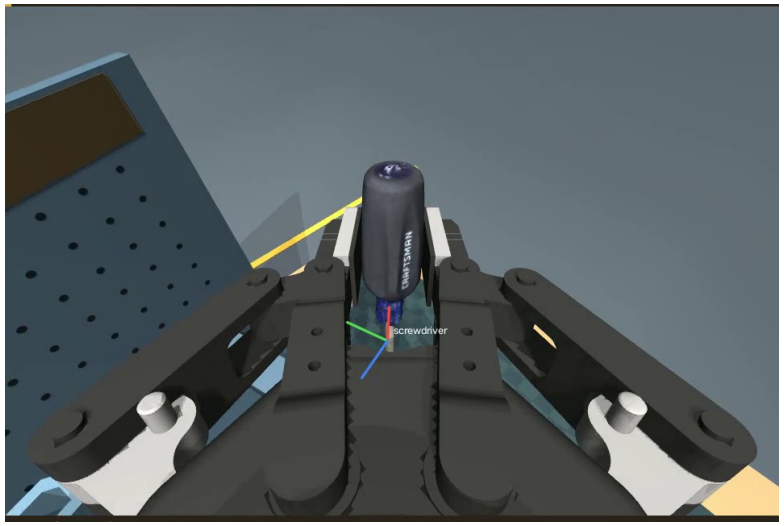
The selected successful trajectory demonstrates feasibility. Low complete-sequence reliability and wide intervals limit conclusions about feedback benefit. Early failures remain part of the system-level outcome.

# Pose accuracy and the physical failure boundary

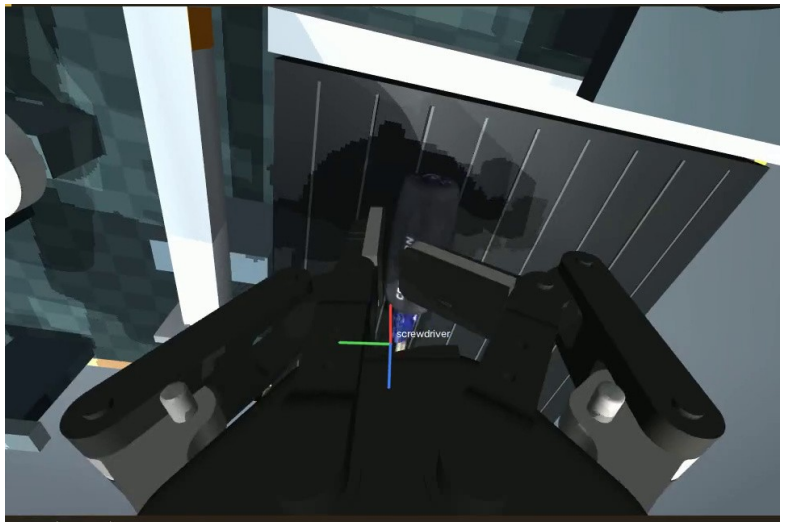


Error decomposition

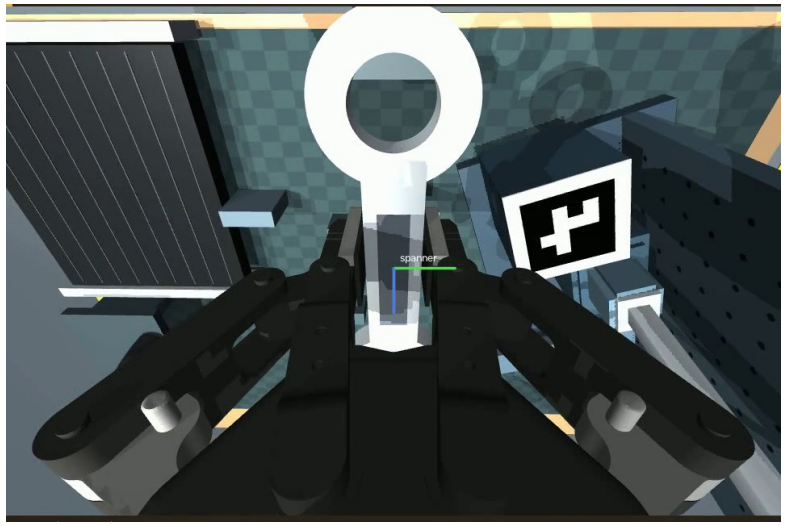
## Visual pose / initial acquisition




## Pickup contact guard




## Grasp reliability after reobservation




| Initial pose metric | Mean |
|---|---|
| Translation / rotation | 2.60 mm / 1.44° |
| ADD / ADD-S | 3.17 / 2.29 mm |

| Failure category | Attempts |
|---|---|
| Pickup contact / hammer planning | 24 / 6 |
| Insertion guard / spanner grasp loss | 6 / 1 |

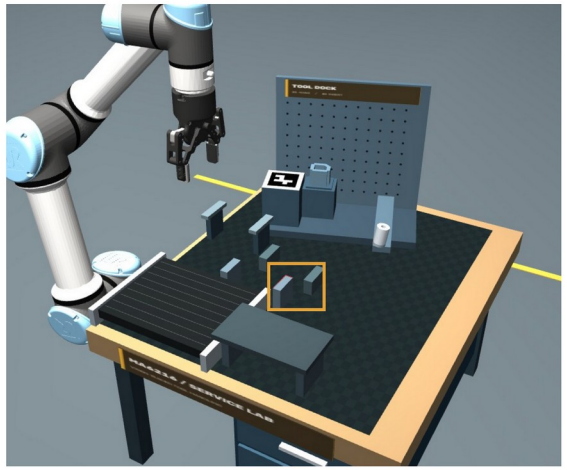
Initial pose error, TCP tracking and final placement are different endpoints. 500 Hz physics is not visual feedback frequency.

# Generalization boundary: background-dependent localization



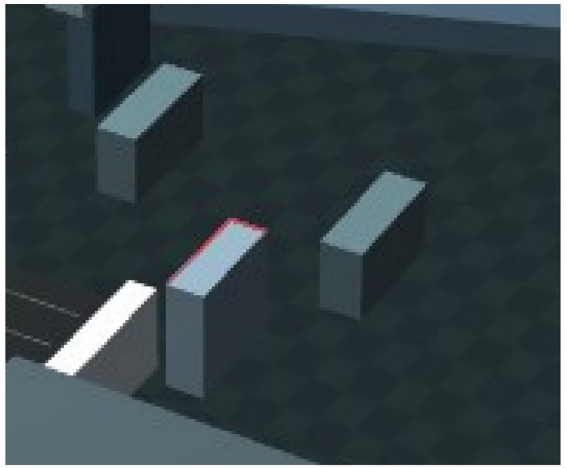


**a** Target-absent scene



Separate background-mismatch diagnosis

**b** Magnified false-positive region



Red pixels: predicted foreground without target

## Observed limitation

Absent rack target, but mismatched background geometry produces a false-positive region.

The zoom is from the same original image.

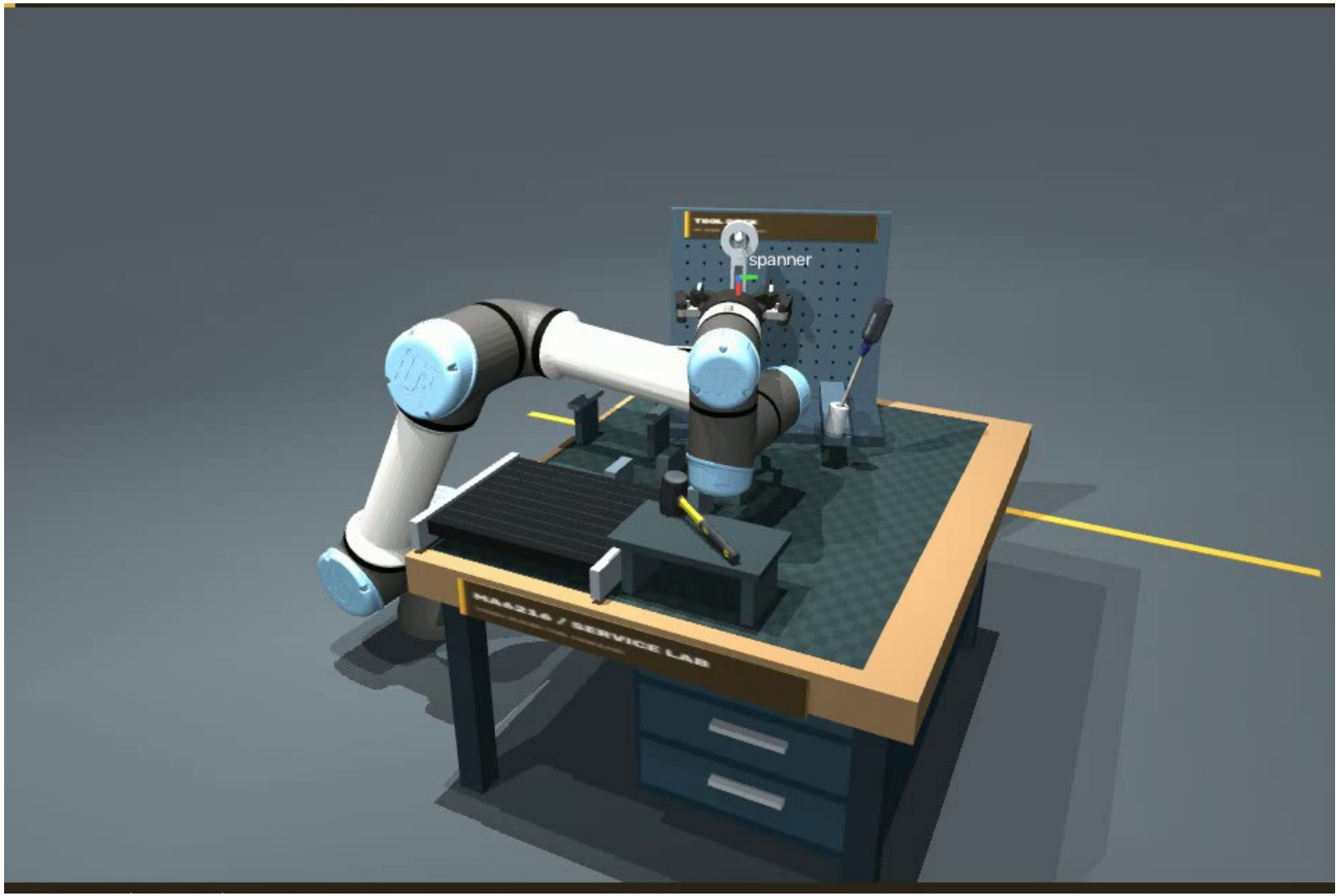
| Assumption | Impact on interpretation |
|---|---|
| Known pickup stations and paired backgrounds | Does not establish arbitrary-clutter recognition |
| Synthetic scans, ideal FK / depth, rigid mounts | Does not establish real-camera or hardware reliability |
| Small reused-seed sample | Broader unseen-scene trials remain necessary |

Separate development diagnosis; not pooled into the 50-scene localization audit or the 48 robot attempts.

# What the integrated system demonstrates



Discussion




## Demonstrated

Independent calibration
Visual 6D pose integration
Physical regrasp and assembly
Auditable feedback comparison

## Next priorities

Improve pickup contacts.
Test changed backgrounds.
Use unseen seeds and external baselines.
Validate on hardware.

High reliability and statistical superiority are not established.

# References and reproducibility



Appendix

Selected methods

Wen et al. FoundationPose. CVPR, 2024.

Zhang. A flexible new technique for camera calibration. TPAMI, 2000.

OpenCV. Camera and hand–eye calibration documentation.

Chaumette & Hutchinson. Visual servo control, Part I. RAM, 2006.

Todorov et al. MuJoCo. IROS, 2012.

Reviewed alternatives

Labbé et al. MegaPose. arXiv:2212.06870, 2022.

Fang et al. GraspNet-1Billion. CVPR, 2020.

Repository evidence

Calibration images + robot FK

Predicted and evaluation masks

48 complete trial summaries

Original successful and failed replays

15 annotated image plates (PDF / SVG)

Frame indices, crop boxes and source hashes

Full references: docs/LITERATURE_AND_SELECTION.md

Experiment report: visual-feedback-calibrated-v1

Simulation images illustrate observed behavior; quantitative conclusions use all predefined attempted trials.